

MLOps Lab 2 worksheet

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1 Aim

The aim of this lab was to train a CNN model on the CIFAR-10 dataset and observe the training behavior using different metrics.

2 Methodology

CIFAR-10 dataset was used for this experiment. I implemented a simple Convolutional Neural Network with two convolution layers using PyTorch. I trained the model for 25 epochs using Adam optimizer and Cross Entropy loss function.

3 Training and Tracking

During training, loss, accuracy, gradient flow and weight updates were observed. Weights and Biases (WandB) was used to track the experiments and visualize the metrics.

4 Observations

- Training loss reduced over time.
- Gradients were stable during training.
- Model training was smooth without major issues.

5 Conclusion

The CNN model was able to learn from the CIFAR-10 dataset. This lab helped in understanding basic MLOps concepts such as experiment tracking and monitoring. the link for visualization is attached along with github and colab link

6 Links

- GitHub Repository: https://github.com/KartikIyer27/MLOps-kartik-m25CSA025/tree/lab2_worksheet
- WandB Project Link: <https://wandb.ai/m25csa025-iit-jodhpur/mlops-lab-cifar10/workspace?nw=nwuserm25csa025>
- Colab Notebook Link: https://colab.research.google.com/drive/1WhlUvZvw5s8h_fVTIDNHKpi2pGDMjuPF#scrollTo=VNvXd0ZcYJld